Practical – 4

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4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using
the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Code:

```
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
series = pd.Series(numbers)
# Grouping by even and odd numbers and calculating the mean
grouped = series.groupby(series % 2 == 0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
print("Mean of even and odd numbers:")
print(grouped)
```

Average time	Maximum time
0.021 s	**0.044 s**
20.67 ms	44.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 44 ms Debug

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd 5.0	Odd 5.0
Even 6.0	Even 6.0
dtype: float64	dtype: float64

Test case 2 13 ms Debug

Expected output	Actual output
4 4 4 6 6 2 2 2 10 12 14 16	4 4 4 6 6 2 2 2 10 12 14 16
Mean of even and odd numbers:	Mean of even and odd numbers:
Even 6.833333	Even 6.833333
dtype: float64	dtype: float64

Test case 3 12 ms Debug

Expected output	Actual output
1 5 7 9 11 13 15 17 111 21 25	1 5 7 9 11 13 15 17 111 21 25
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd 21.363636	Odd 21.363636
dtype: float64	dtype: float64

4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

Code:

```
import pandas as pd

# Provided dictionary of lists
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
}

# Convert the dictionary to a DataFrame
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
print(df)

# Adding a new row
Name = input("New name: ")
Age = int(input("New age: "))
new={'Name':Name,'Age':Age}
```

```
df=df.append(new,ignore_index=True)

# Display the DataFrame after adding a new row
print("After adding a row:\n",df)

# Modifying a row
index = int(input("Index of row to modify: "))
age = int(input("New age: "))
df.at[index, 'Age'] = age



# Display the DataFrame after modifying a row
print("After modifying a row:")
print(df)

# Deleting a row
delt = int(input("Index of row to delete: "))
df = df.drop(delt).reset_index(drop=True)
# Display the DataFrame after deleting a row
print("After deleting a row:")
print(df)

# Adding a new column
gen = input("Enter genders separated by space: ").split()
df['Gender'] = gen

# Display the DataFrame after adding a new column
```

```
print("After adding a new column:")
print(df)

# Modifying a column
df['Name']=df['Name'].str.upper()
# Display the DataFrame after modifying a column
print("After modifying a column:")
print(df)

# Deleting a column
df=df.drop(columns=['Age'])
# Display the DataFrame after deleting a column
print("After deleting a column:")
print(df)
```

Average time	Maximum time
0.183 s	**0.209 s**
183.00 ms	209.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 209 ms Debug

Expected output	Actual output
Original·DataFrame:	Original·DataFrame:
······Name··Age	······Name··Age
0····Alice···25	0····Alice···25
1······Bob···30	1······Bob···30
2··Charlie···35	2··Charlie···35
New·name:·Susan	New·name:·Susan
New·age:·29	New·age:·29
After·adding·a·row:	After·adding·a·row:
·······Name··Age	·······Name··Age
0····Alice···25	0····Alice···25
1······Bob···30	1······Bob···30
2··Charlie···35	2··Charlie···35
3····Susan···29	3····Susan···29
Index·of·row·to·modify:·0	Index·of·row·to·modify:·0
New·age:·27	New·age:·27
After·modifying·a·row:	After·modifying·a·row:
······Name··Age	······Name··Age
0····Alice···27	0····Alice···27
1······Bob···30	1······Bob···30
2··Charlie···35	2··Charlie···35
3····Susan···29	3····Susan···29
Index·of·row·to·delete:·3	Index·of·row·to·delete:·3
After·deleting·a·row:	After·deleting·a·row:
······Name··Age	······Name··Age
0····Alice···27	0····Alice···27
1······Bob···30	1······Bob···30
2··Charlie···35	2··Charlie···35
Enter·genders·separated·by·space:·F M M	Enter·genders·separated·by·space:·F M M
After·adding·a·new·column:	After·adding·a·new·column:
······Name··Age·Gender	······Name··Age·Gender
0····Alice···27······F	0····Alice···27······F
1······Bob···30······M	1······Bob···30······M
2··Charlie···35······M	2··Charlie···35······M

After adding a new column:	After adding a new column:
Name Age Gender	Name Age Gender
0 Alice 27 F	0 Alice 27 F
1 Bob 30 M	1 Bob 30 M
2 Charlie 35 M	2 Charlie 35 M
After modifying a column:	After modifying a column:
Name Age Gender	Name Age Gender
0 ALICE 27 F	0 ALICE 27 F
1 BOB 30 M	1 BOB 30 M
2 CHARLIE 35 M	2 CHARLIE 35 M
After deleting a column:	After deleting a column:
Name Gender	Name Gender
0 ALICE F	0 ALICE F
1 BOB M	1 BOB M
2 CHARLIE M	2 CHARLIE M

4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is **'B'**).

Note:

Refer to the displayed test cases for better understanding.

Code:

```
import pandas as pd

# Read the text file into a DataFrame

file = input()

data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])

print("First five rows:")

print(data.head())

# write your code here..

average_age = round(data["Age"].mean(),2)

print("Average age:",average_age)


filtered_students = data[data["Grade"] <= "B"]

print("Students with a grade up to B")

print(filtered_students)
```

Average time	Maximum time
0.095 s	**0.137 s**
94.50 ms	137.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 137 ms Debug

Expected output	Actual output
studentdata.txt	studentdata.txt
First five rows:	First five rows:
Name Age Grade	Name Age Grade
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A
3 Mike 23 C	3 Mike 23 C
4 Sony 21 B	4 Sony 21 B
Average age: 23.29	Average age: 23.29
Students with a grade up to B	Students with a grade up to B
Name Age Grade	Name Age Grade
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A
4 Sony 21 B	4 Sony 21 B
5 Amia 26 A	5 Amia 26 A

4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

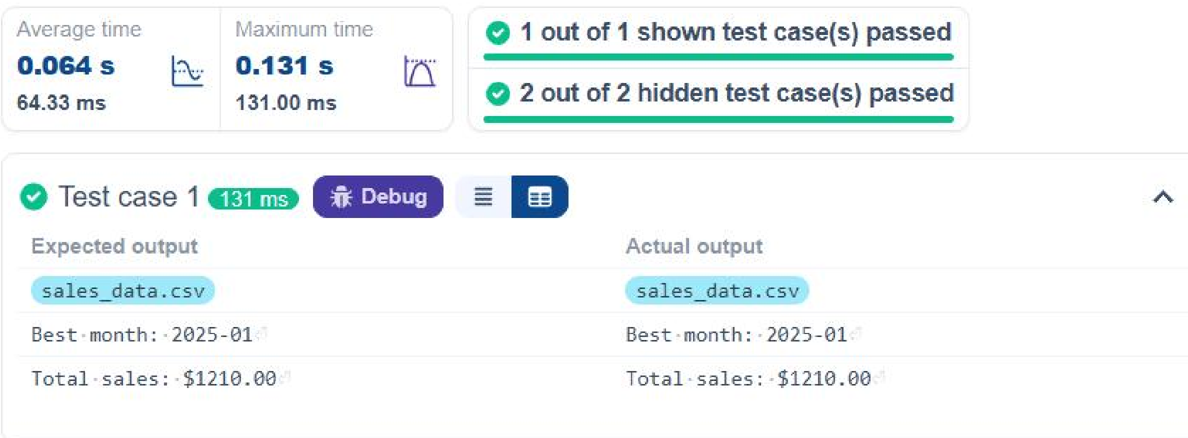
2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)
df['Date']=pd.to_datetime(df['Date'])
df['Month']=df['Date'].dt.to_period('M').astype(str)

# Find the month with the highest total sales
df['Sales']=df['Quantity']*df['Price']
monthly_sales=df.groupby('Month')['Sales'].sum()
best_month =monthly_sales.idxmax()
highest_sales = monthly_sales.max()

print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```



4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

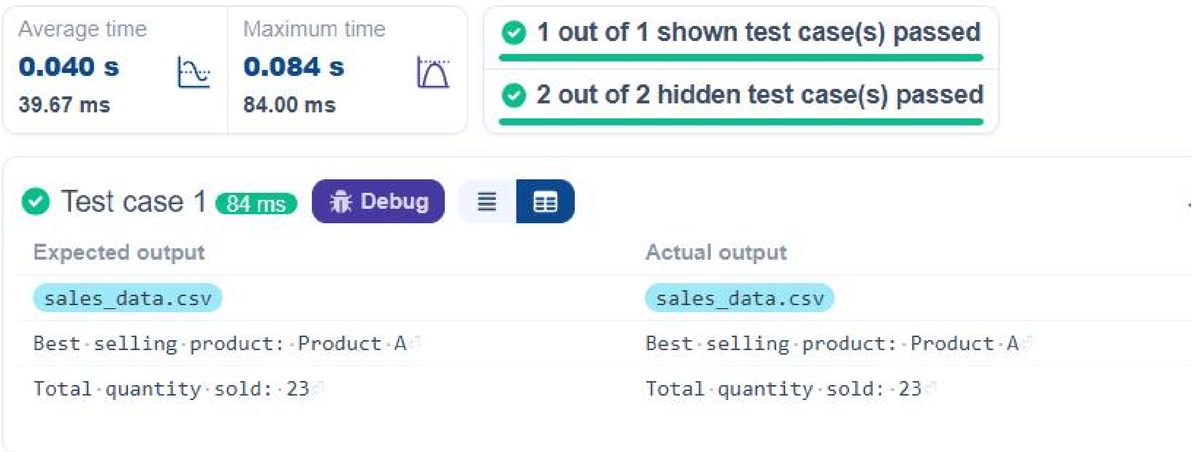
2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

product_sales = df.groupby("Product")["Quantity"].sum()
# Find the product with the highest total quantity sold
best_product = product_sales.idxmax()
highest_quantity = product_sales.max()

# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```



4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

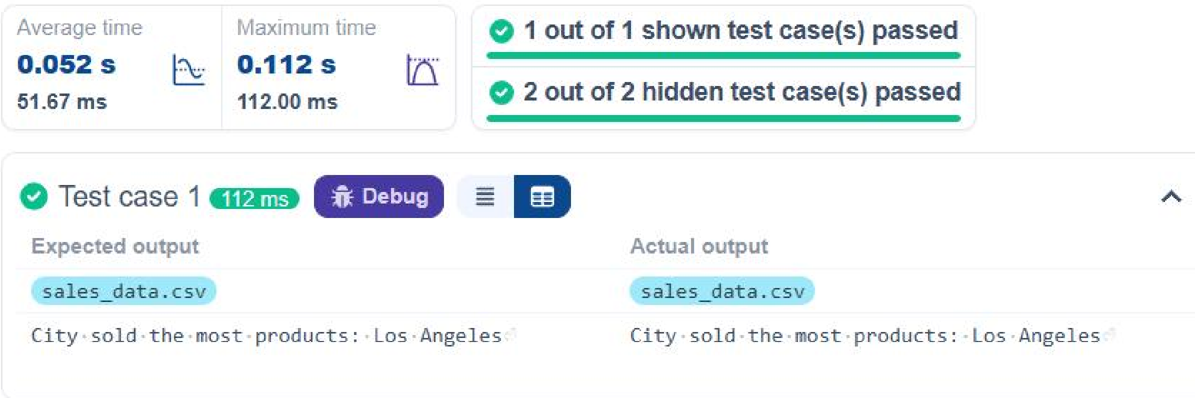
2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

city_sales = df.groupby("City")["Quantity"].sum()
best_city = city_sales.idxmax()

# Display the result
print(f"City sold the most products: {best_city}")
```



4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product **A**,5,20,New York

2025-01-01,Product **B**,3,15,Los Angeles

2025-01-02,Product **A**,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product **B**,2,15,Chicago

2025-01-03,Product **A**,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product **B**,5,15,Los Angeles

2025-01-05,Product **A**,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

Explanation:

Transactions:

- **2025-01-01:** Product A, Product B
- **2025-01-02:** Product A, Product C
- **2025-01-03:** Product B, Product A
- **2025-01-04:** Product C, Product B

- **2025-01-05:** Product A, Product C

Now, let's count how often the pairs of products appear together:

- **Product A and Product B**: Appear in transactions on 2025-01-01 and 2025-01-03.
- **Product A and Product C**: Appear in transactions on 2025-01-02 and 2025-01-05.
- **Product B and Product C**: Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- **Product A and Product B** (2 times)
- **Product A and Product C** (2 times)

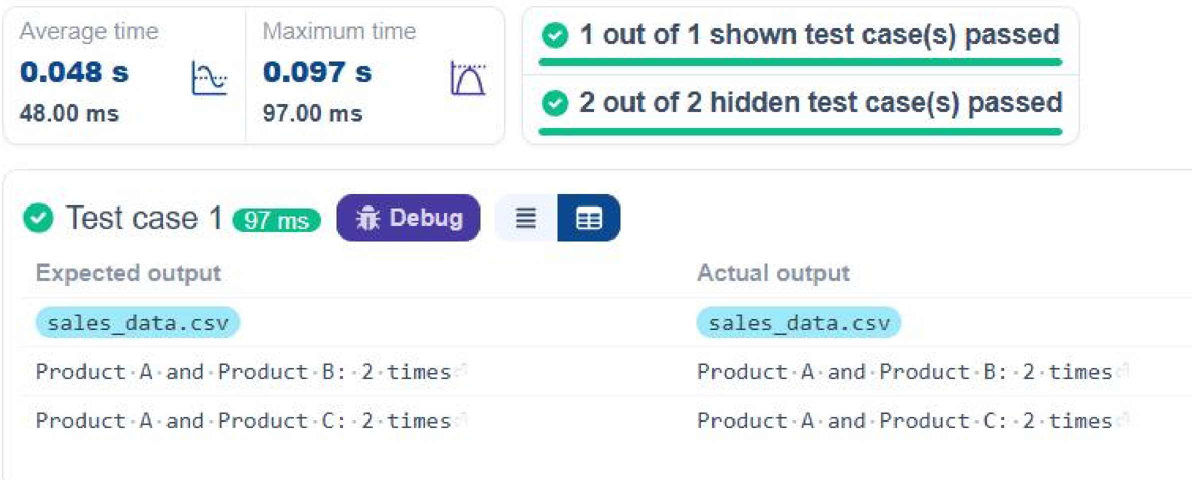
Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```
import pandas as pd

from itertools import combinations

from collections import Counter


# Prompt user to input the file name

file_name = input()


# Read data from the specified CSV file

df = pd.read_csv(file_name)


# write the code

grouped = df.groupby("Date")["Product"].apply(list)
```

```
pairs_counter = Counter()
for products in grouped:
    pairs=combinations(sorted(products), 2)
    pairs_counter.update(pairs)

max_frequency = max(pairs_counter.values())
most_common_pairs = [(pair, freq) for pair, freq in
pairs_counter.items() if freq == max_frequency]

for (product1, product2), frequency in most_common_pairs:
    print(f"{product1} and {product2}: {frequency} times")

# Output the most frequent product pairs
```



4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using .info()).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (mode()).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,
Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# 1. Display the first 5 rows of the dataset

print(data.head())


# 2. Display the last 5 rows of the dataset
```

```
print(data.tail())

# 3. Get the shape of the dataset
print(data.shape)

# 4. Get a summary of the dataset (info)
print(data.info())

# 5. Get basic statistics of the dataset
print(data.describe())

# 6. Check for missing values
print(data.isnull().sum())

# 7. Fill missing values in the ‘Age’ column with the median age
median_age = data['Age'].median()
data['Age'].fillna(median_age, inplace=True)

# 8. Fill missing values in the ‘Embarked’ column with the mode
mode_embarked = data['Embarked'].mode() [0]
data['Embarked'].fillna(mode_embarked, inplace=True)

# 9. Drop the ‘Cabin’ column due to many missing values
data.drop('Cabin', axis=1, inplace=True)

# 10. Create a new column 'FamilySize’ by adding ‘SibSp' and ‘Parch'
data['FamilySize']=data['SibSp']+data['Parch']
```

Average time **0.510 s** 510.00 ms	Maximum time **0.510 s** 510.00 ms

1 out of 1 shown test case(s) passed

Test case 1 510 ms Debug

Expected output	Actual output
PassengerId Survived Pclass ... Fare Cabin Embarked	PassengerId Survived Pclass ... Fare Cabin Embarked
0 1 0 3 ... 7.2500 NaN S	0 1 0 3 ... 7.2500 NaN S
1 2 1 1 ... 71.2833 C85 C	1 2 1 1 ... 71.2833 C85 C
2 3 1 3 ... 7.9250 NaN S	2 3 1 3 ... 7.9250 NaN S
3 4 1 1 ... 53.1000 C123 S	3 4 1 1 ... 53.1000 C123 S
4 5 0 3 ... 8.0500 NaN S	4 5 0 3 ... 8.0500 NaN S
[5 rows x 12 columns]	[5 rows x 12 columns]
PassengerId Survived Pclass ... Fare Cabin Embarked	PassengerId Survived Pclass ... Fare Cabin Embarked
886 887 0 2 ... 13.00 NaN S	886 887 0 2 ... 13.00 NaN S
887 888 1 1 ... 30.00 B42 S	887 888 1 1 ... 30.00 B42 S
888 889 0 3 ... 23.45 NaN S	888 889 0 3 ... 23.45 NaN S
889 890 1 1 ... 30.00 C148 C	889 890 1 1 ... 30.00 C148 C
890 891 0 3 ... 7.75 NaN Q	890 891 0 3 ... 7.75 NaN Q

[5 rows x 12 columns]	[5 rows x 12 columns]
(891, 12)	(891, 12)
<class 'pandas.core.frame.DataFrame'>	<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890	RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):	Data columns (total 12 columns):
# Column Non-Null Count Dtype	# Column Non-Null Count Dtype
--- ------ -------------- -----	--- ------ -------------- -----
0 PassengerId 891 non-null int64	0 PassengerId 891 non-null int64
1 Survived 891 non-null int64	1 Survived 891 non-null int64
2 Pclass 891 non-null int64	2 Pclass 891 non-null int64
3 Name 891 non-null object	3 Name 891 non-null object
4 Sex 891 non-null object	4 Sex 891 non-null object
5 Age 714 non-null float64	5 Age 714 non-null float64
6 SibSp 891 non-null int64	6 SibSp 891 non-null int64
7 Parch 891 non-null int64	7 Parch 891 non-null int64
8 Ticket 891 non-null object	8 Ticket 891 non-null object
9 Fare 891 non-null float64	9 Fare 891 non-null float64
10 Cabin 204 non-null object	10 Cabin 204 non-null object
11 Embarked 889 non-null object	11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)	dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB	memory usage: 66.2+ KB
None	None
PassengerId Survived Pclass ... SibSp Parch Fare	PassengerId Survived Pclass ... SibSp Parch Fare
count 891.000000 891.000000 891.000000 ... 891.000000 891.000000 891.000000	count 891.000000 891.000000 891.000000 ... 891.000000 891.000000 891.000000
mean 446.000000 0.383838 2.308642 ... 0.523008 0.381594 32.204208	mean 446.000000 0.383838 2.308642 ... 0.523008 0.381594 32.204208
std 257.353842 0.486592 0.836071 ... 1.102743 0.806057 49.693429	std 257.353842 0.486592 0.836071 ... 1.102743 0.806057 49.693429
min 1.000000 0.000000 1.000000 ... 0.000000 0.000000 0.000000	min 1.000000 0.000000 1.000000 ... 0.000000 0.000000 0.000000
25% 223.500000 0.000000 2.000000 ... 0.000000 0.000000 7.910400	25% 223.500000 0.000000 2.000000 ... 0.000000 0.000000 7.910400
50% 446.000000 0.000000 3.000000 ... 0.000000 0.000000 14.454200	50% 446.000000 0.000000 3.000000 ... 0.000000 0.000000 14.454200
75% 668.500000 1.000000 3.000000 ... 1.000000 0.000000 31.000000	75% 668.500000 1.000000 3.000000 ... 1.000000 0.000000 31.000000
max 891.000000 1.000000 3.000000 ... 8.000000 6.000000 512.329200	max 891.000000 1.000000 3.000000 ... 8.000000 6.000000 512.329200

[8 rows x 7 columns]	[8 rows x 7 columns]
PassengerId 0	PassengerId 0
Survived 0	Survived 0
Pclass 0	Pclass 0
Name 0	Name 0
Sex 0	Sex 0
Age 177	Age 177
SibSp 0	SibSp 0
Parch 0	Parch 0
Ticket 0	Ticket 0
Fare 0	Fare 0
Cabin 687	Cabin 687
Embarked 2	Embarked 2
dtype: int64	dtype: int64

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

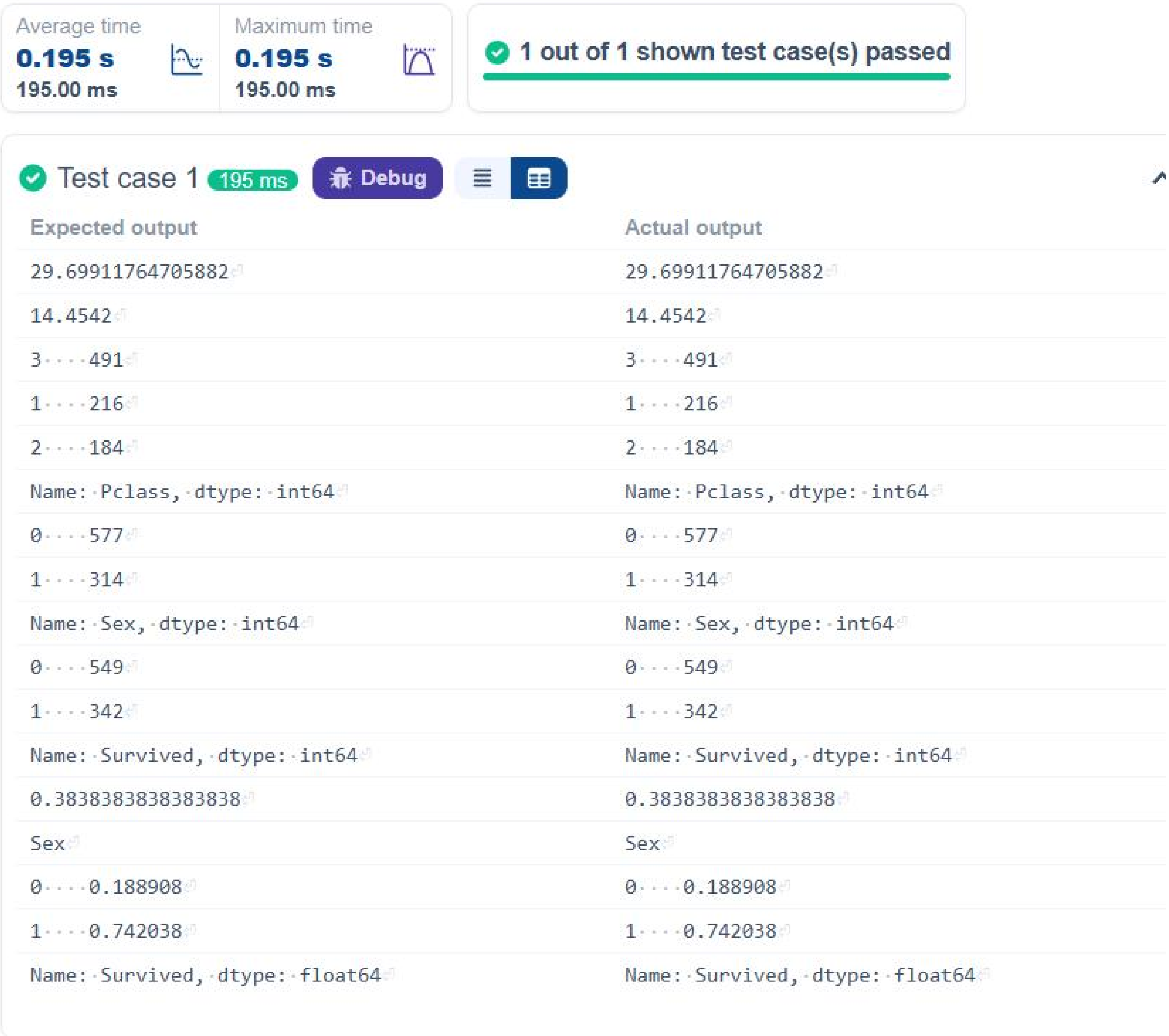
Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']


# 1. Create a new column ‘IsAlone' (1 if alone, 0 otherwise)

data['Alone'] = np.where(data['FamilySize'] == 0, 1, 0)


# 2. Convert ‘Sex' to numeric (male: 0, female: 1)

data['Sex'] = data['Sex'].map({'male' : 0, 'female' : 1})
```

```
# 3. One-hot encode the 'Embarked' column
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 4. Get the mean age of passengers
mean_age = data['Age'].mean()
print(mean_age)


# 5. Get the median fare of passengers
median_fare = data['Fare'].median()
print(median_fare)


# 6. Get the number of passengers by class
passengers_by_class = data['Pclass'].value_counts()
print(passengers_by_class)


# 7. Get the number of passengers by gender
passengers_by_gender = data['Sex'].value_counts().sort_index()
print(passengers_by_gender)


# 8. Get the number of passengers by survival status
passengers_by_survival = data['Survived'].value_counts().sort_index()
print(passengers_by_survival)
# 9. Calculate the survival rate
```

```
survival_rate = data['Survived'].mean()
print(survival_rate)


# 10. Calculate the survival rate by gender
print(data.groupby('Sex')['Survived'].mean())
```



4.2.7. Titanic Dataset Analysis and Data Cleaning – 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,
Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs
Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2.
3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']

data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# 1. Calculate the survival rate by class

print(data.groupby('Pclass')['Survived'].mean())


# 2. Calculate the survival rate by embarked location

print(data.groupby('Embarked_S')['Survived'].mean())
```

```
# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())

# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())

# 5. Get the average fare by class
print(data.groupby('Pclass')['Fare'].mean())

# 6. Get the average age by class
print(data.groupby('Pclass')['Age'].mean())

# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())

# 8. Get the average fare by survival status
print(data.groupby('Survived')['Fare'].mean())

# 9. Get the number of survivors by class
print(data[data['Survived']==1]['Pclass'].value_counts())

# 10. Get the number of non-survivors by class
print(data[data['Survived']==0]['Pclass'].value_counts())
```

Average time

0.220 s

220.00 ms

Maximum time

0.220 s

220.00 ms

1 out of 1 shown test case(s) passed

Test case 1 220 ms Debug

Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 0.506073	0 0.506073
1 0.336957	1 0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0 0.303538	0 0.303538
1 0.552795	1 0.552795
2 0.578431	2 0.578431
3 0.724138	3 0.724138
4 0.200000	4 0.200000
5 0.136364	5 0.136364
6 0.333333	6 0.333333
7 0.000000	7 0.000000
10 0.000000	10 0.000000

Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0 0.505650	0 0.505650
1 0.303538	1 0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1 84.154687	1 84.154687
2 20.662183	2 20.662183
3 13.675550	3 13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1 38.233441	1 38.233441
2 29.877630	2 29.877630
3 25.140620	3 25.140620
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0 30.626179	0 30.626179
1 28.343690	1 28.343690
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0 30.626179	0 30.626179
1 28.343690	1 28.343690
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0 22.117887	0 22.117887
1 48.395408	1 48.395408
Name: Fare, dtype: float64	Name: Fare, dtype: float64
1 136	1 136
3 119	3 119
2 87	2 87
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
3 372	3 372
2 97	2 97
1 80	1 80
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64

4.2.8. Titanic Dataset Analysis and Data Cleaning – 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,
Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs
Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)



# 1. Get the number of survivors by gender

print(data[data['Survived']==1]['Sex'].value_counts())


# 2. Get the number of non-survivors by gender

print(data[data['Survived']==0]['Sex'].value_counts())
```

```
# 3. Get the number of survivors by embarked location
print(data[data['Survived']==1]['Embarked_S'].value_counts())

# 4. Get the number of non-survivors by embarked location
print(data[data['Survived']==0]['Embarked_S'].value_counts())

# 5. Calculate the percentage of children (Age < 18) who survived
children=data[data['Age']<18]
print(children['Survived'].mean())
# 6. Calculate the percentage of adults (Age >= 18) who survived
adults=data[data['Age']>=18]
print(adults['Survived'].mean())

# 7. Get the median age of survivors
print(data[data['Survived']==1]['Age'].median())

# 8. Get the median age of non-survivors
print(data[data['Survived']==1]['Age'].median())

# 9. Get the median fare of survivors
print(data[data['Survived']==1]['Fare'].median())

# 10. Get the median fare of non-survivors
print(data[data['Survived']==0]['Fare'].median())
```

Average time	Maximum time	
0.199 s	**0.199 s**	1 out of 1 shown test case(s) passed
199.00 ms	199.00 ms	

Test case 1 199 ms Debug

Expected output	Actual output
female 233	female 233
male 109	male 109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male 468	male 468
female 81	female 81
Name: Sex, dtype: int64	Name: Sex, dtype: int64
1 217	1 217
0 125	0 125
Name: Embarked_S, dtype: int64	Name: Embarked_S, dtype: int64
1 427	1 427
0 122	0 122
Name: Embarked_S, dtype: int64	Name: Embarked_S, dtype: int64
0.5398230088495575	0.5398230088495575
0.3810316139767055	0.3810316139767055
28.0	28.0
28.0	28.0
26.0	26.0
10.5	10.5